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Enhancing LLM-based clinical reasoning in anesthesiology via graph-augmented retrieval and explainable generation

Meng Wang¹, Yangyang Shen², Bingcheng Zhao³, Xuefeng Zhou⁴, Lan Sun⁴ and Xing Liu^{5*}

Abstract

Purpose: This study aims to enhance the capabilities of large language models (LLMs) in anesthesiology decision support, leveraging a graph-based Retrieval-Augmented Generation (RAG) framework to improve analytical reasoning and deliver evidence-driven results.

Methods: Based on various clinical guidelines and textbooks, we constructed an Anesthesiology Knowledge Graph using UMLS as a reference ontology. We present a graph-based RAG framework, AnesGraph-RAG, which integrates pre-retrieval judgment, hybrid retrieval, and tailored Chain-of-Thought prompting to enhance reasoning performance. Its variant AnesGraph-RAG-UD further incorporates unfamiliarity-driven graph retrieval and structural querying to improve token efficiency.

Results: The constructed anesthesiology knowledge graph comprises 212313 entities and 529845 relations. Evaluation on the expert-annotated dataset confirms its high entity recall (82.19%) and precision (82.13%). Evaluations on the Anesthesiology Attending Physician Qualification Examination dataset indicate that our framework achieves superior performance over ChatGPT-3.5-turbo, ChatGPT-4o, and DeepSeek-V3, with relative improvements of 2.4%, 7.3%, and 3.7% in average accuracy, respectively. The performance gains are particularly pronounced in domains assessing professional knowledge and professional practical skills. In addition, our case study demonstrates that the enhanced post-hoc explanation module offers reasoning paths supported by more detailed evidence.

Conclusion: In conclusion, by integrating structured domain knowledge with adaptive reasoning mechanisms, our graph-based RAG framework improves both the accuracy and interpretability of LLM responses in anesthesiology tasks. These results demonstrate the practical value of combining domain-specific knowledge graphs with LLMs in clinical decision support.

Keywords: Anesthesiology, Knowledge graph, LLM, RAG

Introduction

Annually, more than 300 million surgical operations that need anesthesia are carried out worldwide [1]. Anesthesiologists are required to possess extensive professional knowledge, keen observation, strong analytical skills, and the ability to make rapid decisions. A survey of anesthesiologists' attitudes towards artificial

intelligence (AI) indicated that they have confidence in the ability of AI to complement their work rather than replace it [2].

Recent advancements in AI, especially the rapid development of large language models (LLMs) such as the ChatGPT series [3], show significant potential for clinical decision support. This is due to their ability to understand, generate, and reason about complex medical texts. More than half of healthcare tasks involve evaluating the medical knowledge of LLM, such as answering questions on medical license examinations and making clinical diagnoses [4]. In the

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field of anesthesia, several attempts have recently been made. These include the formulation of preoperative anesthetic plans [5], ASA (American Society of Anesthesiologists) risk stratification [6], and peri-operative complication prediction [7].

However, a major limitation of existing LLMs is that their knowledge is restricted to information within a certain cutoff date. In addition, LLMs are prone to hallucination, generating incorrect or fabricated information, compromising their reliability in high-stakes clinical settings [8]. Although fine-tuning can adapt LLMs to specific domains, it could be computationally expensive and susceptible to catastrophic forgetting. As a more flexible and scalable alternative, Retrieval-Augmented Generation (RAG) enables LLMs to access external knowledge at inference time, reducing hallucinations [9]. Recent studies demonstrated that RAG implementation increased outcomes by 1.35 times compared to baseline LLMs in medical scenarios [10], suggesting the potential of RAG in anesthesiology.

A range of knowledge retrieval strategies has been developed for RAG systems. Notably, graph-based RAG demonstrates an advantage over text RAG due to the richer contextual information provided by knowledge graphs [11]. However, realizing its full potential for high-stakes applications such as clinical decision support systems in Anesthesiology, requires overcoming several significant challenges.

- *Absence of Anesthesiology Knowledge Graphs.* Building large-scale, high-quality, domain-specific knowledge graphs is inherently costly and labor-intensive. This is a major impediment to the broader adoption of Graph RAG. Currently, to the best of our knowledge, no research has explored RAG approaches specifically grounded in anesthesiology-specific knowledge graphs, representing a notable gap in the medical field.
- *Insufficient Domain Adaptation.* Existing graph-based RAG methods may not fully exploit the rich contextual information in the anesthesiology knowledge graphs. For the retrieval, constructing context solely based on entity similarity may omit the complex, multifactorial knowledge required in anesthesiology, leading to suboptimal performance. For the generation, domain-aware reasoning capabilities and token efficiency remain important areas for further optimization.
- *Lack of Reasoning Interpretability.* Most RAG frameworks offer only implicit or fuzzy references to external knowledge that they retrieved before generation, resulting in a lack of interpretable citations for their entire reasoning processes that involve broader

knowledge. This is particularly problematic in anesthesiology clinical decision-making, where transparent and evidence-based reasoning is essential.

These limitations underscore the need for advanced, explainable graph-based RAG frameworks in anesthesiology scenarios. Addressing these issues may help policymakers and clinicians integrate LLMs into clinical workflows more effectively.

To address these gaps, we develop the foundational algorithms for a clinical decision support system in anesthesiology, leveraging a comprehensive anesthesiology knowledge graph and advanced RAG techniques. Our work makes the following key contributions:

- We constructed a comprehensive Anesthesiology Knowledge Graph, the first of its kind, by processing a diverse corpus of clinical guidelines and textbooks. This addresses a practical gap by creating a structured knowledge base for the anesthesiology domain, and it was made possible through an automated pipeline that circumvents the prohibitive cost and labor of manual construction.
- We design and implement two graph-based RAG frameworks, AnesGraph-RAG and AnesGraph-RAG-UD, specifically tailored for anesthesiology graph RAG. AnesGraph-RAG incorporates an improved retrieval-generation pipeline, a pre-judgment mechanism for retrieval necessity, a hybrid retrieval strategy, and tailored Chain-of-Thought (CoT) prompting to enhance accuracy and reduce hallucinations. AnesGraph-RAG-UD further addresses resource constraints with a Familiarity-Aware Graph Retrieval strategy and structural querying for token efficiency.
- We propose an Enhanced Post-Hoc Explanation module within our RAG framework. This addresses the challenge of a lack of reasoning interpretability in complex medical reasoning by providing verifiable citations for key reasoning steps, thereby significantly increasing the transparency and trustworthiness of model responses.

Through extensive experiments, we empirically demonstrate that our graph extraction achieves high precision (82.13%) and recall (82.19%), and our framework surpasses the standard ChatGPT-4o with a relative accuracy improvement of 7.3% on Anesthesiology Attending Physician Qualification Examination datasets. The most notable improvement is observed on the Professional Practical Skills Question Bank, where our method yields a relative accuracy improvement of 13.4%.

Preliminaries

Probabilistic Language Modeling in Large Language Models. Large Language Models (LLMs) [12, 13], have become the foundation of modern natural language processing due to their remarkable ability to generate coherent, context-aware text. These models are typically built upon the transformer architecture [14] and trained on massive corpora using self-supervised learning objectives. The core training task is known as *probabilistic language modeling*, where the model learns to assign probabilities to sequences of text in a way that reflects natural language usage.

Formally, given a discrete vocabulary $\mathcal{V}$ and a token sequence $x = (x_1, x_2, \dots, x_T)$, where $x_t \in \mathcal{V}$, an LLM defines a probability distribution over the entire sequence by factorizing it into conditional probabilities:

$$P_{\theta}(x) = \prod_{t=1}^T P_{\theta}(x_t | x_{<t}), \quad (1)$$

where $x_{<t} = (x_1, \dots, x_{t-1})$ denotes the preceding context and θ represents the model parameters. As observed in works such as [15, 16], the performance of these models improves predictably with increased scale of model parameters, data, and computation. Moreover, purely through probabilistic modeling, LLMs exhibit emergent behaviors such as few-shot generalization and instruction

following, despite no explicit supervision for such tasks during training.

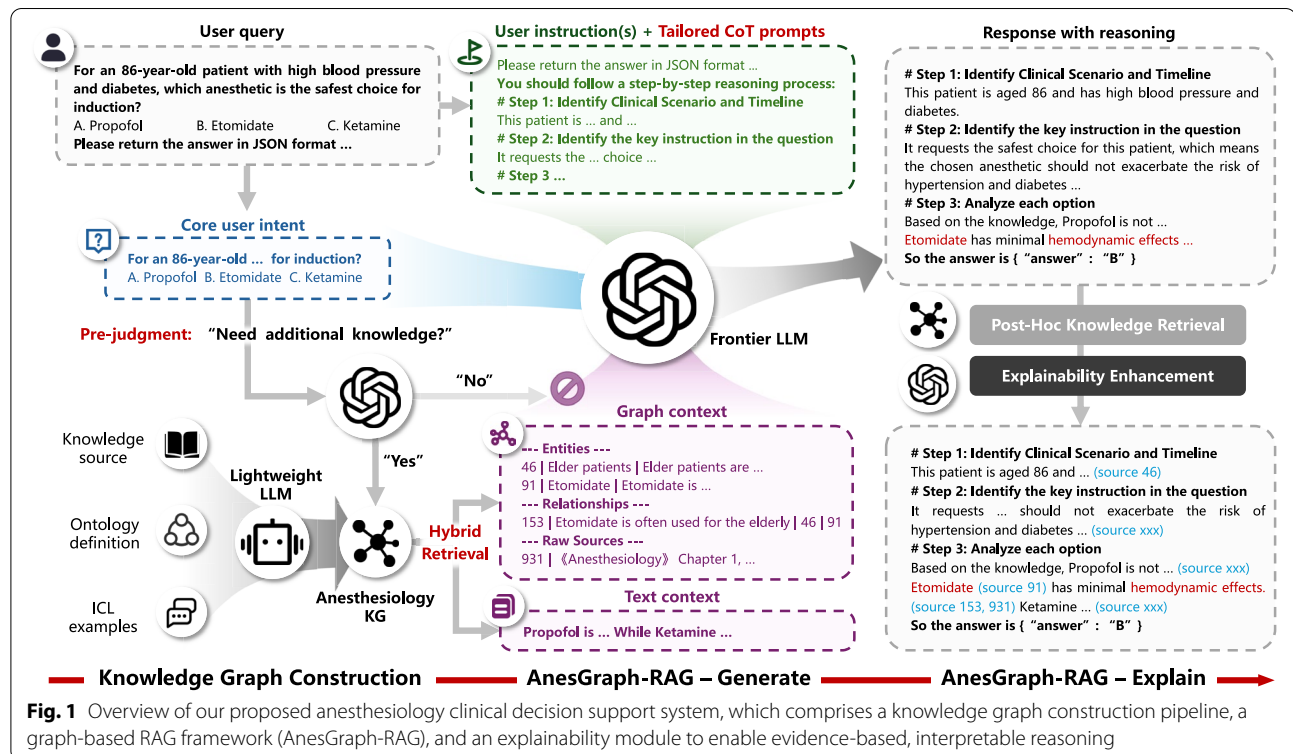
Methodology

Overview

As illustrated in Fig. 1, our goal is to develop a clinical decision support system for anesthesiology, which consists of three core components. To support graph-based RAG algorithms, we first use an automated pipeline powered by a local large-language model to construct a comprehensive knowledge graph of the Anesthesiology domain. This graph is derived by extracting entities and relationships from domain-specific corpora, including clinical guidelines, textbooks, and peer-reviewed literature. Leveraging this knowledge graph, we design a Retrieval-Augmented Generation (RAG) framework, named AnesGraph-RAG, to enable an evidence-based and clinically reliable anesthesiology assistant. To enhance explainability, we further integrate an explainability enhancement module that provides a transparent and traceable reasoning process for RAG.

Anesthesiology knowledge graph construction

The performance of a Retrieval-Augmented Generation (RAG) system is indeed highly dependent on the quality of its knowledge base [17]. To support clinical assistance in anesthesiology, we leverage an LLM-based automated



pipeline to construct a high-quality anesthesiology knowledge graph.

Data sources

To ensure that the domain-specific knowledge graph contains comprehensive knowledge about anesthesiology, its construction is grounded in two major data sources: Anesthesiology-related Textbooks and Anesthesiology-related Clinical Guidelines.

Structured knowledge extraction is grounded in a comprehensive bilingual corpus comprising 5 authoritative anesthesia-related textbooks and 35 clinical practice guidelines from Chinese and English sources. These materials cover general medical sciences (e.g., physiology, pathology), core clinical disciplines (e.g., internal medicine, surgery), and specialized anesthesiology, such as Yao & Artusio's Anesthesiology. The clinical guidelines address key areas including perioperative complications, pharmacologic management, emergency procedures, and care for high-risk patients. Detailed Textbooks and Clinical Guidelines are listed in Table 3 of the Appendix A.

Ontology definitions

To standardize the definition of biomedical entities and relationships in our knowledge graph, we use the Metathesaurus component of the Unified Medical Language System (UMLS) [18] as the reference for entity types and relation types. UMLS was developed by the U.S. National Library of Medicine (NLM). It is widely used in biomedical informatics due to its comprehensive integration of medical terminologies and semantic standards. The Metathesaurus aggregates concepts and relationships from numerous source vocabularies and assigns unified identifiers and semantic types, making it well-suited for ensuring consistency, interoperability, and reusability across biomedical knowledge systems. In our framework, the Metathesaurus serves as a well-defined ontology reference. The detailed entity types and relationship types are presented in Table 4 of Appendix A.

Knowledge extraction scheme

To address the inherent complexity of anesthesia-related data—characterized by its multi-source, heterogeneous, and temporally dynamic nature—we design a unified knowledge extraction framework powered by large language models (LLMs).

At the input layer, we incorporate the mentioned large-scale anesthesiology corpus along with predefined entity and relationship schemas. At the model layer, a lightweight LLM (Doubao 1.5 lite [19]) is employed to process this anesthesia corpus comprising approximately 20 million tokens. To facilitate entity disambiguation and enhance similarity-based retrieval, we further utilize the

BGE-M3 embedding model [20]. The documents are first segmented into small chunks, and the corresponding embedding vectors are then computed. Based on the defined ontology, relevant entities and relationships are extracted to construct the initial knowledge graph, which is subsequently enriched through community detection and community summarization.

Enhancing Extraction Performance via In-Context Learning. To overcome the scarcity of labeled data—common in specialized medical domains such as anesthesiology—we adopt In-Context Learning (ICL) for few-shot knowledge extraction. ICL enables the model to generalize to unseen instances by conditioning on a limited number of annotated examples without requiring parameter updates. Specifically, each ICL prompt includes a set of example pairs (x_i, y_i) , where x_i is the input text and y_i is its corresponding structured output. These examples form a demonstration context $c = (x_1, y_1, \dots, x_k, y_k)$, which is concatenated with a new input x , which is a new unseen sentence/paragraph requiring knowledge extraction, to form the final input (c, x) . The model then generates the target output y according to the conditional probability $P_\theta(y|c, x)$. With well-crafted prompts, the output y typically comprises serialized structured knowledge, such as:

```
((Adverse drug reactions: nausea, respiratory depression),
→ (Disease symptoms: abdominal pain, fever), (Anesthetic
→ events: hypoxemia (time: induction phase) (severity:
→ severe)))
```

We curated ICL examples in collaboration with anesthesiology experts to support the extraction of various domain-specific knowledge categories pertinent to anesthesia scenarios. For detailed prompt templates used in the construction of knowledge graphs, refer to the supplementary material in the Appendix D.

AnesGraph-RAG

Based on our constructed anesthesiology knowledge graph, we further develop a graph-based RAG framework for anesthesiology scenarios. With the structured medical knowledge, our approach reduces hallucinations and improves the accuracy and explainability of generated responses.

Specifically, we propose Anesthesiology Graph RAG (AnesGraph-RAG), a retrieval-augmented generation (RAG) framework for the anesthesiology domain. AnesGraph-RAG is designed to provide clinical decision support by refining the graph-based retrieval and generation pipeline, providing accurate reasoning processes with more detailed evidence. Key innovations of AnesGraph-RAG include an optimized retrieval-generation pipeline, a pre-retrieval judgment mechanism to determine the

necessity of retrieval, a hybrid retrieval strategy for richer contextual grounding, tailored Chain-of-Thought (CoT) prompting to guide stepwise clinical reasoning, and an enhanced post-hoc explanation module to improve interpretability and trustworthiness of the generated responses.

Refined retrieval and generation pipeline

We observe that in real-world clinical decision support systems dialogues, user queries often contain additional user instructions that do not require external knowledge retrieval (e.g., formatting requests or display preferences). These parts may result in low-quality context and thus harm the generation. Hence, we explicitly split the user query into two distinct input fields, rather than directly adopting the full user query for retrieval as in Microsoft's GraphRAG [11]. Specifically, we segment the user input into two components: the core user intent (retrieval target) and auxiliary instructions. Retrieval is performed exclusively on the retrieval target using our anesthesiology knowledge graph to extract relevant context. During the generation stage, the instructions are re-appended to the query as suffixes to guide the LLM generation. Compared to rewriter-based strategies for general-purpose dialogue systems, this simple yet effective retrieval and generation pipeline helps reduce inference latency and mitigates the risk of retrieving overly broad contexts due to rewriter instability.

We embed the retrieval target q using a sentence embedding model and retrieve semantically similar entity descriptions e_i from the knowledge graph via cosine similarity:

$$\text{sim}(q, e_i) = \frac{g(q)^\top g(e_i)}{|g(q)| \cdot |g(e_i)|}, \quad (2)$$

where $g(\cdot)$ denotes the embedding function. To enrich the retrieved context, we follow GraphRAG's strategy by expanding the top- k matched entities with multi-level graph-structured information, including entity attributes, entity relations, linked text sources, and community-level insights derived from graph topology.

Pre-judgment mechanism

Given the strong prior knowledge encoded in modern LLMs through large-scale pretraining and fine-tuning via SFT and RLHF, retrieval is not always a necessity for all queries. We notice that frontier LLMs already have basic knowledge of anesthesiology to some extent. To avoid unnecessary overhead, we introduce a pre-judgment mechanism that evaluates whether a given query necessitates external retrieval. Instead of introducing additional training to embed special tokens to mark the status [21],

we leverage the natural language understanding capabilities of modern LLMs to directly perform this pre-judgment. As illustrated in Fig. 2, the LLM first assesses the difficulty of the user intent and its own confidence in generating an accurate response without additional context. If confident, it proceeds to address the query directly. Otherwise, the model initiates the retrieval stage. This mechanism reduces latency and token usage while maintaining response quality.

Hybrid retrieval mechanism

While graph-based retrieval benefits from the rich context provided by structured knowledge, it can still fail to retrieve relevant information, as the similarity comparison relies on entity descriptions generated by large language models during graph construction, which may be incomplete or overly abstract. Thus, solely retrieving the similar entities and expanding the context within the subgraph may lead to a related but useless context.

To address this issue, we leverage a hybrid retrieval strategy. With graph-based retrieval remaining the primary mechanism, it is supplemented with direct text chunk retrieval from the original corpus used to construct the graph (i.e., the source text units stored in the knowledge graph). Specifically, we compute the similarity between the retrieval target q and each text chunk t_i using $\text{sim}(q, t_i)$, as in Eq. (2). The top- k most relevant text chunks are integrated into the final context to supplement the graph-based context.

Tailored chain-of-thought prompting

To enhance the model's reasoning ability in complex clinical scenarios, we employ Chain-of-Thought (CoT) prompting [22], which enables step-by-step inference. We append a single representative query–reasoning–response example that reflects typical analytical patterns in the anesthesiology domain. It is worth noting that our CoT prompt intentionally avoids overly detailed examples. Instead, it offers a high-level abstraction of the

```
===== Pre-judgment =====
Question: {question}
Based on your understanding of Anesthesiology, are you
confident that you can answer this question *without*
retrieving additional knowledge from the knowledge base?
Respond with:
'yes' – if you can answer it confidently without any additional
information.
'no' – if you would prefer to retrieve more information
before answering.
Please respond with either 'yes' or 'no'.
```

Fig. 2 The prompt template for pre-judgment

reasoning process, omitting task-specific solutions. This aligns with real-world scenarios, where anesthesiology experts, when confronted with new problem categories, can sketch a general reasoning framework based on domain expertise, without exhaustive references to textbooks or clinical guidelines.

Thanks to the explicit separation between user intent and user instructions in our query design, this CoT example can be seamlessly integrated directly after the user instructions, without interfering with the retrieval of core user intent. This helps the model follow multi-step reasoning paths and generate more interpretable and accurate responses. An illustration of our CoT example used in our prompt is shown in Fig. 3.

Crucially, CoT prompting not only improves standalone performance but, when properly designed, also synergizes with the retrieved knowledge in RAG. The step-by-step demonstration encourages the LLM to better integrate and apply the external information retrieved, resulting in more accurate and interpretable answers. This alignment between retrieval and reasoning significantly enhances the overall efficacy of AnesGraph-RAG.

Enhanced post-hoc explanation

One significant advantage that comes with Retrieval-Augmented Generation (RAG) systems, beyond the mitigation of model-induced hallucinations, is the support for post-hoc explainability. By prompting the LLMs to cite retrieved passages alongside the generated response. This enhances transparency by allowing users to trace the provenance of the model's claims and thereby assess their reliability.

However, the standard method of directly requesting LLMs to cite sources often falls short in domains that demand complex, multi-step reasoning, such as anesthesiology case analysis. In such scenarios, key steps in the model's reasoning may involve steps that are not directly supported by retrieved knowledge, leading to imprecise or missing citations. For instance, a clinical query may present a set of vital signs and lab results without explicitly naming a suspected disease. The LLM might correctly infer the disease, treating it as a pivotal step in its reasoning chain. But due to the weak semantic link between the raw clinical data and the inferred diagnosis, the disease name may not be in the retrieved documents. Hence, this crucial inference step remains uncited, resulting in a gap in explainability.

Moreover, two other practical challenges emerge when attempting to enforce stricter citation. First, mandating rigorous citation via prompt engineering, especially within long and complex contexts, can degrade model performance. The Large Language Model (LLM) may

==== Chain-of-Thought example =====

Question:

A 45-year-old male patient with chronic renal failure underwent allograft kidney transplantation under general anesthesia. Upon entering the operating room, his blood pressure was 170/100 mmHg. After anesthesia induction, ..., blood pressure dropped to 90/60 mmHg. **Aramine, Dopamine, and accelerated fluid replacement were administered**, with a central venous pressure of 16 cm H₂O, but there was still no significant increase in blood pressure. The cause of hypotension is _____

- A: Insufficient blood volume
- B: Inadequate dosage of Aramine and Dopamine
- C: Poor cardiac function
- D: Acidosis causing myocardial depression and decreased sensitivity of the myocardium to catecholamines
- E: Norepinephrine has a better pressor effect

Answer:

(Reasoning) I should take a step-by-step thinking process to arrive at the correct answer.

Deconstruct the Clinical Scenario and Timeline

I must break down the patient's information and the sequence of events.

- **<patient information>** (basic conditions, medical history, etc.) is
- **<timeline>** (the sequence of events/operations) is

Identify the key instruction in the question

There are tried methods to raise the patient's blood pressure, they excluded some reasons for hypotension, and ask for the remaining reason (under this Clinical Scenario).

Analyze each choice

Considering **<the patient's condition, the sequence of events, and the medical knowledge >**:

- A is incorrect because
- B is incorrect because
- C is incorrect because
- D is correct because
- E is incorrect because

So the answer is:

{"answer": "E"}

Fig. 3 An example used in our chain-of-thought prompt

struggle to adhere perfectly to all user instructions, leading to a decline in the quality and accuracy of the response. Second, a default policy of appending citations to all queries, while seemingly thorough, is inefficient. It leads to unnecessary token consumption in cases where users do not require detailed explanations.

To address these limitations, we propose an Enhanced Post-Hoc Explanation mechanism. In our two-stage process, the system first generates and delivers the response to the user's query. Concurrently, it logs the complete, unfiltered output from the LLM, including the detailed chain-of-thought reasoning that led to the initial answer.

Subsequently, if the user requests an explanation for a specific response, the system initiates the second stage. It retrieves the original query and the corresponding logged reasoning process using the same graph retrieval function (Post-Hoc Knowledge Retrieval). Despite

some methods also attempting to improve the citation quality after the generation, they fix the citation based on the response and original retrieved documents [23], which can not provide citations that were not included in the original retrieval. We perform a more detailed secondary search on the knowledge graph, which may cover omitted entities/relationships in the first retrieval. The LLM is then prompted to augment the original reasoning path with citations drawn from these newly retrieved, more relevant sources. This ensures maximal citation coverage for the key nodes in the reasoning chain, thereby providing a comprehensive and verifiable explanation. This method also allows for flexible prompting strategies, enabling the model to present its reasoning in preferred formats to best suit the user's needs. An example template for such a post-hoc explanation prompt is provided in Fig. 5.

AnesGraph-RAG-UD

To address the challenges of resource-constrained scenarios, we also propose a more token-efficient Graph RAG method. Unlike conventional RAG approaches that broadly retrieve related knowledge from graphs or texts for every query, our method takes the increasingly powerful intrinsic capabilities of modern large language models (LLMs) into account. Going beyond the simple pre-judgment mechanism adopted in AnesGraph-RAG, we further introduce a Familiarity-Aware Graph Retrieval variant that selectively targets anesthesiology knowledge that the LLM is less familiar with. This approach is instantiated in the form of the Unfamiliarity-Driven Anesthesiology Graph RAG (AnesGraph-RAG-UD).

AnesGraph-RAG-UD begins with a self-assessment phase, in which the LLM is prompted to identify the core topic and key concepts within the query, and to rank these concepts by its own familiarity (prompt template shown in Fig. 6). This step ensures that concept selection prioritizes relevance while avoiding redundancy with the intrinsic knowledge of the LLM.

To manage retrieval efficiency, we set a fixed context budget—the maximum number of concepts permitted in the final context. Based on the self-assessed familiarity ranking, we iteratively exclude the most familiar concepts, ensuring that the final context comprises only the most relevant and least familiar information needed to answer the query effectively. Instead of traditional embedding-based retrieval, we employ structured querying with Cypher, a declarative graph query language designed for flexible and precise graph traversal. We design a Cypher query template that retrieves exactly the linked nodes corresponding to the required concepts,

enabling fast and accurate knowledge graph retrieval for AnesGraph-RAG-UD (detailed in Fig. 7).

After constructing the tailored context, we use the same Chain-of-Thought (CoT) prompting approach as in AnesGraph-RAG to generate the final answer.

Experiments

In this section, we first present the implementation details of the knowledge graph construction and the evaluation of our RAG-based framework (Sect. 4.1). We then report experimental results on two fronts: evaluating the quality of the constructed knowledge graph (Sect. 4.2) and assessing clinical reasoning performance via anesthesiology QA tasks (Sect. 4.3). Ablation studies are conducted to verify the effectiveness of each module in AnesGraph-RAG (Sect. 4.4), and finally, we demonstrate our enhanced post-hoc explanation through a representative case study (Sect. 4.5).

Implementation details

Knowledge graph

The detailed data sources, entities&relationship definitions are in Appendix A. The main model used for knowledge graph construction is Doubao 1.5-lite [19], which inherits the cutting-edge Mixture-of-Experts (MoE) sparse-activation architecture and several optimizations designed for large-scale deployment. For detailed prompts used for knowledge graph construction, we provide a supplementary file, see Appendix D.

RAG evaluation

To evaluate our AnesGraph-RAG, we construct the QA test dataset using three categories of questions from the Anesthesiology Attending Physician Qualification Examination: the Basic Knowledge Question Bank (BK), Professional Practical Skills Question Bank (PPS), and Professional Knowledge Question Bank (PK). A distinguishing feature of the PPS question bank is its focus on clinical scenario-based questions. From each category, we randomly sample 100 multiple-choice questions and translate them into English QA pairs using ChatGPT-4o [3] as the translator (translate prompt is shown in Fig. 8). Since all questions are in multiple-choice format, accuracy is used as the primary evaluation metric.

To comprehensively assess the performance of the proposed method, we employ multiple base models: HuatuoGPT-o1(7B) [24], ChatGPT-3.5-turbo, and ChatGPT-4o [3]. All models are configured with a temperature of 0.5 and utilize the same result parsing strategy. HuatuoGPT-o1 is included as a representative medical LLM baseline. Its selection is based on the substantial computational resources required for training and fine-tuning larger models, making an 8-billion-parameter

medical LLM a more representative choice. The results for HuatuoGPT-o1 and ChatGPT-3.5-turbo are averaged over five runs, those for ChatGPT-4o are averaged over two runs.

For baseline comparisons, as there are no known Anesthesiology Knowledge Graphs at this scale to the best of our knowledge, we consider two primary categories of methods:

- *Standard LLMs*: these models operate without any retrieval augmentation, relying solely on their internal knowledge for reasoning and response generation. They serve as a reference to highlight the importance of incorporating external knowledge for complex medical reasoning. We use the question with an output format notation as the input.
- *Text-RAG LLMs*: these models integrate traditional retrieval-augmented generation (RAG) mechanisms. Specifically, we use BGE-m3 [20] as the embedding model and LanceDB as the vector index. The source text is segmented into chunks of 300 tokens each. For each query, the top 5 most similar chunks (based on embedding similarity) are retrieved as the context. Model inputs are updated in 'context-question' format. This setup helps assess how conventional text-based retrieval can improve the reasoning performance of LLMs, and will reveal how the knowledge graph boosts the performance.

Our proposed method includes two variants (as detailed in Sect. 3.3), which integrate graph-structured knowledge into the QA process in different ways. Their detailed settings are:

- *Shared Settings*: The implementation of our method employs the same embedding model as the Text-RAG LLMs baseline, namely BGE-m3 [20], with the same hyperparameters. In detail, the chunk size is set to 300 tokens, and the embedding dimension is 1024. For inference-related hyperparameters, we set the maximum tokens to 8191, maximum retries to 20, and request timeout to 7200 s.
- *AnesGraph-RAG*: For the retrieval of graph knowledge, the overall token limit is 3000 tokens, where the number of entities and the number of relationships are both limited to 10. When hybrid retrieval is enabled, we ensure that the combined token consumption of graph and textual retrieval does not exceed 3000 tokens. Specifically, the graph retrieval budget is reduced to 1000 tokens, while textual retrieval uses the top-5 chunks.
- *AnesGraph-RAG-UD*: There is a maximum try of 3 for LLM to extract the concepts in the question. The

dynamic context length is restricted to between 2 and 10 items (fewer than 3000 tokens); the one with the fewest items will be chosen if a varying number of concepts are valid.

Evaluating knowledge graph quality

The final constructed knowledge graph contains 212313 entities and 529845 relationships. To rigorously assess the performance of the LLM in entity extraction within the anesthesiology domain, as well as evaluating the quality of the constructed knowledge graph, we selected five paragraphs at random from *Miller's Anesthesia*, a widely recognized authoritative textbook in anesthesiology. These paragraphs were chosen because they contain rich domain-specific knowledge and a diverse array of entity types, with the total number of distinct entity categories exceeding 500. This ensures a comprehensive evaluation that reflects the complexity and variability of real-world clinical text.

The annotation process involved two human annotators: one domain expert with formal training in anesthesiology and one non-expert with no professional medical background. Both annotators independently labeled all entities in the selected paragraphs, strictly adhering to a predefined ontology of medical entity types. To ensure consistency and correctness, all annotations were subsequently reviewed and validated by a senior clinician with extensive experience in both anesthesiology and medical informatics. By involving experts in a multi-step annotation procedure, we ensured high-quality annotations that are highly reliable for benchmarking.

We then compared the entities extracted by the LLM against this manually annotated benchmark dataset. Evaluation was conducted using standard information extraction metrics: precision (P), recall (R), and their harmonic mean, the F1 score. The metrics are formally defined as follows:

$$\begin{aligned} \text{Precision (P)} &= \frac{TP}{TP + FP}, \\ \text{Recall (R)} &= \frac{TP}{TP + FN}, \\ \text{F1 Score} &= \frac{2 \cdot P \cdot R}{P + R}, \end{aligned} \quad (3)$$

where TP (True Positives) is the number of entities correctly identified by the model (i.e., both entity type and span match the manually annotated benchmark dataset), FP (False Positives) is the number of entities extracted by the model that do not match any entity in the manually annotated benchmark dataset, FN (False Negatives) is

the number of entities present in the manually annotated benchmark dataset but missed by the model. As visualized in Fig. 4, the model achieved a precision of 82.13%, a recall of 82.19%, resulting in an F1 score of 82.16%. These results demonstrate that the LLM is capable of identifying a substantial proportion of relevant medical entities with high accuracy.

Evaluating clinical reasoning via anesthesiology QA tasks

To evaluate the model’s capability in logical reasoning within medical thinking, we adopted multiple-choice questions from the Anesthesiology Attending Physician Qualification Examination Datasets as the test sets. The performance of our AnesGraph-RAG and

Table 1 Performance of our proposed methods, AnesGraph-RAG and AnesGraph-RAG-UD, compared with standard LLMs and Text-RAG-enhanced LLMs on three anesthesiology test sets: the Basic Knowledge Question Bank (BK), Professional Practical Skills Question Bank (PPS), and Professional Knowledge Question Bank (PK)

Model	Method	BK	PPS	PK	Average
HuatuoGPT-o1	Standard	44.60	34.00	31.00	36.53
ChatGPT-3.5-turbo	Standard	54.80	57.00	53.40	55.07
	Text-RAG	57.00	53.40	54.40	54.93
	AnesGraph-RAG	53.20	59.80	56.20	56.40
ChatGPT-4o	Standard	72.00	63.50	70.50	68.67
	Text-RAG	71.50	67.00	70.50	69.67
	AnesGraph-RAG	73.00	72.00	73.00	72.67
	AnesGraph-RAG-UD	76.00	71.50	73.00	73.67
DeepSeek-V3	Standard	79.00	76.00	72.00	75.67
	Text-RAG	76.00	75.50	73.00	74.83
	AnesGraph-RAG	80.50	77.00	78.00	78.50
	AnesGraph-RAG-UD	79.00	76.50	76.50	77.33

Accuracy (%) is reported. Our approaches achieve the highest average accuracy across all test sets, surpassing standard ChatGPT-3.5-turbo by 2.4%, ChatGPT-4o by 7.3%, and DeepSeek-V3 by 3.7%. Notably, on ChatGPT-4o, the largest gain occurs on PPS (professional practical skills), with a 13.4% improvement. For DeepSeek-V3, the improvement is relatively modest, as the baseline accuracy is already high

AnesGraph-RAG-UD on three kinds of anesthesiology test sets (described in Sect. 4.1) is presented in Table 1, compared against standard LLMs and Text-RAG LLMs.

Compared with standard LLMs, our methods achieve the highest average accuracy across all test sets, as quantified by the relative improvement metric, defined as (B–A)/A. Specifically, they surpass ChatGPT-3.5-turbo by 2.4%, ChatGPT-4o by 7.3%, and DeepSeek-V3 by 3.7%. The most notable gain is observed in the PPS (Professional Practical Skills) task when using ChatGPT-4o as the backbone model, where our approach delivers a 13.4% performance improvement. This strong performance highlights the significant advantages of our constructed knowledge graph and RAG pipeline in tackling anesthesiology-related questions, particularly in enabling complex reasoning over both professional knowledge and practical skill scenarios. These benefits arise from the high quality of our carefully designed knowledge graph, which provides a reliable and comprehensive knowledge base, together with the RAG pipeline, which effectively pinpoints the required information and guides the LLM’s reasoning process.

We observe that the number of parameters in the foundational LLM for RAG methodologies critically influences the magnitude of improvement over their corresponding standard LLMs. Specifically, smaller models exhibit a reduced capacity to effectively integrate contextual information retrieved, which consequently curtails the performance benefits derived from RAG. Additionally, these models frequently display limitations in instruction-following, failing to produce outputs in the correct JSON format consistently. On the other hand, larger models such as DeepSeek-V3, which are trained on extensive corpora, already exhibit high baseline accuracy, leading to diminishing marginal gains. While the improvements may appear moderate, our method enables the model to achieve both the highest average accuracy and the best performance across all datasets.

Ablation study

To rigorously evaluate the individual contribution of each component within our proposed AnesGraph-RAG framework, we conducted a comprehensive ablation

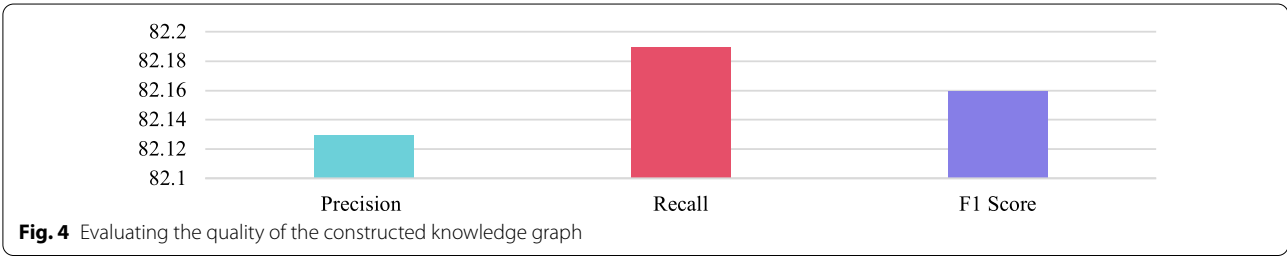


Table 2 The ablation study of our AnesGraph-RAG method on three anesthesiology test sets using ChatGPT-4o as the base LLM

Model	Method	BK	PPS	PK	Average
ChatGPT-4o	Standard	72.00	63.50	70.50	68.67
	Ablation 1	66.50	63.00	70.00	66.50
	Ablation 2	69.50	67.50	70.50	69.17
	Ablation 3	70.50	70.50	68.50	69.83
	Ablation 4	74.00	67.50	67.00	69.50
	AnesGraph-RAG	73.00	72.00	73.00	72.67

We detail a four-level ablation, starting from a standard LLM and incrementally adding our key components, the graph retrieval, pre-judgment module, hybrid retrieval, and Chain-of-Thought (CoT) prompting, to arrive at the full AnesGraph-RAG model. The results demonstrate that each component positively contributes to performance

study. The results, summarized in Table 2, systematically deconstruct the performance of our model on three distinct anesthesiology test sets. Our analysis follows a six-stage, incremental approach, beginning with a baseline configuration that uses the standard ChatGPT-4o model. From this baseline, we sequentially integrate the core modules of our design:

- *Ablation 1*: Standard LLM + our implemented GraphRAG. Instead of adopting the original open-source GraphRAG strategy, we redesigned the process to further decouple retrieval and answer generation. See Sect. 3.3 for details.
- *Ablation 2*: Standard LLM + our GraphRAG + Pre-judge. Based on *Ablation 1*, we enable the LLM to autonomously decide whether knowledge retrieval is necessary.
- *Ablation 3*: Standard LLM + our GraphRAG + Pre-judge + Hybrid Retrieval. Building on *Ablation 2*, we incorporate additional text retrieved via Text-RAG during the retrieval phase, rather than relying solely on knowledge graph retrieval.
- *Ablation 4*: Standard LLM + our GraphRAG + Pre-judge + Hybrid Retrieval + ICL. This setting most closely approximates AnesGraph-RAG. On top of *Ablation 3*, we include QA examples in the prompt but without structuring them as detailed step-by-step Chain-of-Thought demonstrations.

The findings demonstrate that each component provides a tangible performance enhancement. The introduction of the pre-judgment and hybrid retrieval modules each yields positive gains over the baseline, confirming their utility. However, the most critical insight arises from the interaction between the components. While the stand-alone performance improvement from the hybrid retrieval module is moderate, we observe a significant synergistic effect when it is combined with the CoT prompting. This suggests that simply providing the model with retrieved knowledge is insufficient; the CoT framework is crucial as it guides the model to reason over, synthesize, and effectively utilize this external information. This synergy allows the model to unlock the full potential of the retrieved context, leading to the substantial performance boost observed in the final, fully-integrated AnesGraph-RAG model and validating our architectural design choices.

Case study

To demonstrate the explainability of our proposed AnesGraph-RAG method and highlight its strong question-answering capabilities, we conduct a case study on the Professional Practical Skills Question Bank—the most representative and also the most challenging task in our dataset. This task demands both extensive domain-specific knowledge and advanced reasoning abilities. The interpretability of its reasoning process is thus crucial for evaluating the practical value and reliability of our method.

As shown in Tbox 1, the selected case was about a 77-year-old patient who was scheduled for laparoscopic cholecystectomy. The patient complicated multiple chronic diseases, such as hypertension, chronic left ventricular dysfunction, bronchial asthma and bronchitis. In terms of anesthesia airway assessment, this patient may have a potentially difficult airway risk because of small mandible and habitual snoring. The question focuses on selecting the most appropriate strategy for the induction of anesthesia using propofol. This question mainly tests the understanding of anesthesia induction in elderly patients with chronic diseases, which is a topic combining drug application with anesthesia risk assessment.

Tbox 1: Question

Question:

Patient, male, 77 years old, 87 kg, 165 cm, is scheduled for laparoscopic cholecystectomy due to cholecystitis and gallstones. The patient has a history of bronchitis and bronchial asthma for many years and has been hospitalized for repeated chest tightness and discomfort in the precordial area. An electrocardiogram showed left ventricular hypertrophy, ST segment depression in leads II and III, and inverted T waves. A cardiac ultrasound revealed reduced motion of the left ventricular inferior wall and an ejection fraction of 50%. Preoperative blood pressure was 160/98 mmHg, and the patient did not take antihypertensive medication regularly. The patient snored habitually during sleep and has a small mandible. If propofol is used for induction, the correct approach is:

A: When using propofol target-controlled infusion (TCI), the initial target plasma concentration should be set at 4 µg/ml

B: A stepwise TCI can be set, starting with an initial plasma target concentration of 1 µg/ml

C: For a single propofol induction, the dose can be given at 2 to 2.5 mg/kg

D: To alleviate the patient's anxiety, midazolam 3 to 5 mg can be given, followed by propofol 2 to 2.5 mg/kg for induction

E: All of the above are acceptable.

Correct Answer: B.

To enhance post-hoc explainability, AnesGraph-RAG revisits our anesthesiology knowledge graph to retrieve more targeted and contextually relevant knowledge that supports the reasoning process. As shown in Tbox 2, the post-hoc knowledge retrieval results contain multiple pieces of domain knowledge that are directly pertinent to the clinical scenario described in the question.

For instance, the retrieved evidence includes:

- Descriptions of appropriate propofol induction dosages for general patient populations (e.g., 1.5–2.5 mg/kg in healthy adults), which serve as a reference point for evaluating their applicability to elderly patients with comorbidities (Entity 38789).
- Explanations of target-controlled infusion (TCI), emphasizing its pharmacokinetic basis and its role in maintaining anesthesia with greater precision and stability (Relationship 337209, 337207)—a method especially relevant for patients with cardiovascular risks.
- Clinical insights such as the warning from Source 27600: “rapid bolus dose administration of

propofol during procedural sedation may result in undesirable cardiorespiratory depression, especially in the elderly.” This directly supports the conclusion that standard bolus induction may not be safe for this patient.

These retrieved items map closely to the key decision points in the original question—how to balance sedation effectiveness with safety concerns in an elderly patient with compromised cardiac and respiratory function. By highlighting the risks of bolus administration and the benefits of a gradual TCI approach, the retrieved knowledge provides essential support for selecting Option B as the correct and safest answer.

Tbox 2: Post-Hoc Knowledge Retrieval

—Entities—

id|entity|description|number of relationships

38789|Healthy, Unpremedicated Adults|Healthy, Unpremedicated Adults are a group of patients for whom propofol dosage and administration information is provided. A specific dose of propofol (1.5 - 2.5 mg kg⁻¹) is required to induce anaesthesia in this group of people. |2

...

—Relationships—

id|source|target|description|weight|rank|links

337209|TCI|TCI of Propofol|TCI of Propofol is a specific type of TCI for maintenance of anaesthesia|9.0|13|1

337207|TCI|pharmacokinetic data and equations|Target - controlled infusion (TCI) techniques use pharmacokinetic data and equations|9.0|8|1

...

—Text sources—

id|text

14883|C.3. Would you use propofol for induction of anesthesia? Propofol may be the induction agent of choice for the hemodynamically stable patient with reactive airways. In comparison to induction with thiopental, thiamylal, or methohexital, 2.5 mg per kg propofol induction resulted in a significantly lower incidence of wheezing after tracheal intubation in asthmatic patients. The incidence of wheezing was 0%, 45%, and 26% in patients who received propofol, a thiobarbiturate, and oxybarbiturate, respectively. Another study in unselected patients found a significantly lower...

27600|However, the use of propofol may be associated with serious and potentially fatal side-effects. These include cardiovascular effects such as hypotension and bradycardia, and respiratory effects such as hypoventilation, hypoxaemia and apnoea. In addition, at similar depths of sedation, propofol is more likely than midazolam to cause airway obstruction, due to upper airway collapsibility, suggesting that particular vulnerability exists after transition from conscious to unconscious sedation.1 Rapid bolus dose administration of propofol during procedural sedation may result in undesirable cardiorespiratory depression, especially in the elderly.

...

Based on the re-retrieved evidence, AnesGraph-RAG generates a structured, step-by-step reasoning chain, as shown in Tbox 3. Each reasoning step is explicitly supported by citations to corresponding graph nodes or text sources, enabling transparent validation of clinical judgments:

- In Step 1, the system identifies key clinical constraints—cardiovascular instability and respiratory vulnerability—which are common in elderly patients with asthma and left ventricular dysfunction (Source 27600).
- Step 2 contextualizes propofol's use, recognizing its lower incidence of airway hyperreactivity compared to barbiturates (Source 14883).
- In Step 3–4, different induction modalities are evaluated: Target-Controlled Infusion (TCI) is preferred for its titratability and controlled pharmacokinetics (Source 337209), while bolus-based single-dose induction is cautioned against due to abrupt hemodynamic shifts, especially in this population (Source 38789).
- Finally, Step 5 concludes that the optimal approach is a stepwise TCI induction starting at 1 µg/ml (Option B), as it aligns with both pharmacological principles and patient-specific risk mitigation (Source 337207, 27600).

This traceable reasoning process not only enhances answer reliability but also reinforces the interpretability of large language model (LLM)-assisted medical reasoning. By mapping each decision point to structured or textual knowledge, AnesGraph-RAG ensures that domain experts can verify, trust, and potentially refine the automated response.

Tbox 3: Explainable Reasoning Process

```
// Detailed reasoning with proper citation.
To ... . Let's break down the reasoning process step by step.
Step 1: Identify the Clinical Problem and Urgency. The patient ... cardiovascular issues ... respiratory concerns ... . The clinical problem is to safely induce anesthesia for laparoscopic cholecystectomy, minimizing risks of cardiovascular and respiratory complications (Source 27600).
Step 2: Define the Immediate Therapeutic Goal. The immediate goal is to ... maintains appropriate sedation while minimizing hemodynamic instability and respiratory depression. Propofol is considered due to its rapid onset and favorable profile in terms of airway reactivity (Source 14883).
Step 3: Enumerate Viable Therapeutic Categories Based on the Situation. Considering the patient's age and cardiac status, we must choose an induction method that allows careful titration:
Target-Controlled Infusion (TCI): Allows precise control of propofol plasma concentrations, potentially reducing cardiovascular impact (Source 337209).
Single Dose Induction: Typically involves a bolus dose, which may cause abrupt changes in hemodynamics (Source 38789).
Step 4: Evaluate Each Category Against Clinical Constraints.
TCI Approach: Starting ... , can provide a gradual increase in sedation, reducing the risk of hypotension and cardiovascular depression (Source 337207). This is important given the patient's reduced cardiac function and hypertension.
Single Dose Induction: A dose of 2 to 2.5 mg/kg could be excessive for an elderly patient with cardiac issues, leading to potential cardiovascular compromise (Source 38789).
Step 5: Justify the Optimal Choice. Given the patient's ... is the safest option, leading to the answer B. This decision is based on the ability of TCI to provide controlled sedation with reduced risk of cardiovascular depression, crucial for this patient's profile (Source 27600, 337209).
```

Related works

Clinical applications of LLMs

Large Language Models (LLMs) demonstrate diverse clinical applications, prominently in aiding the diagnostic process by evaluating a wide range of clinical data to generate diagnoses swiftly [25–30]. Studies have shown their diagnostic potential across various specialties [31–35]. Beyond diagnostics, LLMs augment physician knowledge by summarizing literature [36] and providing specialized information to non-experts [32, 33, 37], thereby streamlining initial evaluations [37, 38] and patient triage [39, 40]. They also support clinical decision-making by offering rapid access to research [25], treatment guidelines, and drug

information [26, 27, 29], aiding in chronic disease management and personalized care strategies [36]. LLMs have the potential for continuous patient monitoring as well, alerting providers to early warning signs [27, 29, 37, 41], although concerns about inaccurate referencing persist [42–45]. In addition, LLMs can significantly reduce administrative burdens by automating tasks such as note-writing [39], scheduling [46], drafting correspondence [29], and managing insurance preauthorizations [25], ultimately decreasing physician workload and burnout [30] and allowing for more direct patient care.

Despite these broad advancements, limited attention has been given to the application of LLMs in anesthesiology. While a few studies have evaluated LLM performance in anesthesiology [47, 48] or proposed fine-tuned domain-specific models [49], they rarely address the hallucination challenges that may arise in real-world anesthetic decision-making. In contrast, our work focuses on the integration of a knowledge graph-based retrieval-augmented generation (RAG) framework tailored for anesthesiology. This approach offers a more reliable and scalable solution for clinical decision support in this critical domain.

Knowledge graph-enhanced LLMs and RAG

From a technical perspective, recent studies have focused on improving the reliability and factual grounding of LLM outputs, with Retrieval-Augmented Generation (RAG) emerging as a widely adopted approach [9, 50–54]. The integration of structured knowledge, particularly through knowledge graphs, has shown promise in enhancing contextual reasoning and reducing hallucinations [55–60]. Compared to unstructured document repositories [61], knowledge graphs provide structured, semantically rich, and inferable information, making them particularly well-suited for augmenting LLMs and RAG systems [55–60]. Several works [62–67], have further proposed training sequence-to-sequence models from scratch, where dialogue generation is conditioned on entities extracted from knowledge graphs. Building upon the core RAG concept, Microsoft's GraphRAG [11] further enhances complex reasoning by constructing and leveraging knowledge graphs for information retrieval.

However, constructing a domain-specific knowledge graph is often labor-intensive, and the LLM-based knowledge graph construction pipeline still needs major modifications to suit the anesthesiology domain. Furthermore, RAG approaches that rely solely on graph indexing often fail to fully leverage the rich contextual relationships embedded within domain-specific

knowledge graphs, resulting in suboptimal performance. Moreover, common solutions enforce the LLMs to explicitly incorporate retrieved knowledge during generation through prompt engineering, which may compromise the overall generation quality. Additionally, newly inferred entities that emerge during the reasoning process may be absent from the initial retrieval results, leading to reduced interpretability of the final response.

Discussion

Technical innovations and advantages

In this work, we construct a domain-specific knowledge graph for anesthesiology and enhance the conventional graph-based RAG (Retrieval-Augmented Generation) framework to deliver robust clinical assistance. This contributes significantly to the development of a more comprehensive anesthesiology clinical decision support system.

Guided by expert knowledge, we design the graph's data sources and ontology definitions, and leverage a local LLM to automate the construction process. Our entity extraction pipeline achieves a precision of 82.13% and a recall of 82.19%.

To improve the RAG process, in AnesGraph-RAG, we introduce a series of retrieval optimization strategies that enhance both the efficiency and quality of retrieved information. Additionally, we develop a domain-specific Tailored Chain-of-Thought (CoT) prompting technique, enabling the model to better utilize the retrieved knowledge in reasoning. Furthermore, our Enhanced Post-hoc Explanation module conducts a second-pass graph-based retrieval to further improve the explainability of the graph-RAG framework.

Empirical findings and challenges

Extensive evaluations demonstrate the effectiveness of our proposed approach. Across three challenging anesthesiology benchmarks, AnesGraph-RAG and its variant AnesGraph-RAG-UD consistently outperform both standard and text-RAG-enhanced large language models. In particular, our models achieve a 13.4% improvement over ChatGPT-4o on the Professional Practical Skills (PPS) questions, underscoring their superior performance in complex scenario-based reasoning tasks.

Ablation studies confirm that each component of AnesGraph-RAG contributes meaningfully to the overall performance, with a powerful synergy observed between hybrid retrieval and CoT prompting. These results emphasize the importance of not only retrieving relevant content but also effectively incorporating it into multi-step reasoning.

While AnesGraph-RAG-UD exhibits improved average performance and better token efficiency, the use of dynamic context length control via structural queries may compromise real-time responsiveness, especially when operating on large-scale graphs. Therefore, we recommend AnesGraph-RAG as a more general and adaptable RAG solution.

Limitations and future works

Despite the promising results, our study has several limitations that we leave for future exploration.

Regarding the construction of the knowledge graph, we employed relatively small-scale local models. We anticipate that leveraging more powerful local language models, combined with enhanced data cleaning procedures, could lead to a higher-quality graph with further improved coverage and accuracy. In addition, while our automated pipeline supports incremental updating by seamlessly incorporating new corpora—such as textbooks or clinical guidelines—more substantial modifications to the KG (e.g., deletion or alteration of existing knowledge) remain challenging. This limitation is beyond the current scope of our work, but we recognize it as an important direction for future research.

In terms of prompt engineering, the current designs of prompts for knowledge extraction and chain-of-thought (CoT) prompting in the RAG framework still require some domain expertise. A potential direction for future work is to investigate the automatic generation of domain-specific prompts via fine-tuned domain-adaptive language models, thereby reducing manual intervention.

As for our enhanced post-hoc explanation method, the post-hoc knowledge retrieval process is guided by the model's intermediate reasoning trace, which helps target more contextually relevant knowledge. We hypothesize that if the system conducts several rounds of iterative

“try-to-answer and reason, then refine retrieval targets” before issuing a final response, it could further improve the precision of the retrieved knowledge and the overall answer quality. However, we observed that this potential strategy introduces latency and incurs token consumption. Therefore, we leave this promising enhancement for future work.

Conclusion

In this study, targeting the development of clinical decision support systems in anesthesiology, we construct a domain-specific knowledge graph and present AnesGraph-RAG and AnesGraph-RAG-UD—two novel RAG frameworks enhanced by this graph to facilitate complex anesthesiology decision-making. By introducing innovations in retrieval, reasoning, and explainability, our approach achieves significant improvements in both accuracy and interpretability over existing LLM and RAG baselines. These findings underscore the potential of integrating structured domain knowledge with adaptive reasoning mechanisms to enhance the effectiveness of LLM-based clinical assistants in specialized fields such as anesthesiology.

Appendix A: Detailed data sources for the knowledge graph construction

An overview of the main sources utilized in the construction of the Anesthesiology Knowledge Graph is presented in Table 3. These 40 sources consist of 5 textbooks and 35 clinical guidelines.

Table 4 presents the entity and relationship types in UMLS.

Table 3 Overview of Textbooks and Clinical Guidelines used for the construction of Anesthesiology Knowledge Graph

Textbooks	
ANESTESIOLOGIA_Textbook of Regional anesthesia	Pre-Hospital Anesthesia Handbook
Pinnock Fundamentals of Anaesthesia	Yao - Yao & Artusio's Anesthesiology_Problem-Oriented Patient Management-LWW (2016)
Textbook of Anaesthesia	
Clinical Guidelines	
2016 ESA Guidelines: Management of Severe Perioperative Bleeding	AIT Consensus Statement_Intensive treatment of children over 3 years of age with general anesthesia: Part 1
2017 ACFAS Clinical Consensus Statement: Perioperative Management	AIT Consensus Statement_Intensive treatment of children over 3 years of age with general anesthesia: Part II
2017 Guidelines for perioperative smoking management	AIT_Consensus Statement on Outpatient Anesthesia
2018 CAS Opinion: Sedation Procedures	AIT_Nutritional Therapy in the Pediatric Intensive Care Unit
2019 APSIC Guidelines: Prevention of Surgical Site Infection	Anaesthesia - 2018 - White - International Fragility Fracture Network Delphi Consensus Statement on Principles of Anesthetic Management for Patients with Hip Fracture
2019 ERAS Guidelines: Perioperative Management in Gynecology and Obstetrics/Oncology	Anaesthesia - 2021 - White - Principles of environmentally?sustainable anaesthesia a global consensus statement from the
2020 ICSI Health Care Guidelines: Perioperative Period (6th Edition)	Anaesthesia - 2023 - Kelly - Implementing human factors in anaesthesia guidance for clinician departments and hospitals
2021 JSA Practice Guidelines: Perioperative Allergic Reactions	Anaesthesia - 2023 - Kinsella - Handling injectable medications in anaesthesia
2014 AATS Guidelines: Prevention and Management of Atrial Fibrillation and Flutter during Perioperative Thoracic Surgery	Anaesthesia - 2023 - Wiles - Position statement from the Editors of Anaesthesia and Anaesthesia Reports on best practice
2022+ERAS Recommendations: Guidelines for perioperative care in liver surgery	JCM 2022 Multinational expert consensus statement on total intravenous anesthesia (TIVA) using the Delphi method
2022+ERAS Guidelines: Perioperative Care for Liver Transplantation	2022 Opioid Use Disorder in Anesthesia and Critical Care: Prevention, Diagnosis, and Management
2023+ESAIC/ESTES consensus guidelines: Recognition, treatment and prevention of perioperative cardiac arrest	2012 NHS guidelines for perioperative management of adults with diabetes
2023+ESAIC Evidence-Based Consensus Guidelines: Postoperative Delirium in Adult Patients (Updated Version)	Non-anesthesiologists should not be allowed to use propofol for procedural sedation: a consensus statement from 21 European national anesthesia societies
2023+ China Evidence-Based Guidelines: Prevention and Management of Perioperative Infection	2047-0525-2-12
2019 Consensus Guidelines of the Society of Anesthesia and Perioperative Nursing for Oncology on the Perioperative Management of Patients Undergoing Cytoreductive Surgery and Hyperthermic Intraperitoneal Chemotherapy (CRS-HIPEC)	Prognostically relevant myocardial injury and inflammation in patients undergoing percutaneous coronary intervention
European Neuromuscular Centre consensus statement on anaesthesia in patients with neuromuscular disorders	COVID-19 safety: aerosol-generating procedures and cardiothoracic surgery and anaesthesia — Australian and New Zealand consensus statement
Ventilating multiple patients on a single ventilator: Statement from the French Society of Anaesthesia and Intensive Care (SFAR)	The rationale for the recommendations of the European Paediatric Fasting Guideline: Improving paediatric anaesthesia and perioperative medicine
IJA-Consensus statement on anesthesia for day care surgery	

Table 4 The entity types and relationship types in UMLS

Type	List
Entities	Entity, Physical Object, Anatomical Structure, Manufactured Object, Clinical Drug, Substance, Conceptual Entity, Idea or Concept, Finding, Laboratory or Test Result, Sign or Symptom, Organism Attribute, Intellectual Product, Language, Occupation or Discipline, Organization, Group Attribute, Group, Event, Activity, Occupational Activity, Health Care Activity, Laboratory Procedure, Diagnostic Procedure, Therapeutic or Preventive Procedure, Research Activity, Governmental or Regulatory Activity, Educational Activity, Phenomenon or Process, Human-caused Phenomenon or Process, Natural Phenomenon or Process, Biologic Function, Physiologic Function, Pathologic Function, Disease or Syndrome, Cell or Molecular Dysfunction, Experimental Model of Disease, Injury or Poisoning
Relationships	associated_with, physically_related_to, part_of, consists_of, contains, connected_to, interconnects, branch_of, tributary_of, ingredient_of, spatially_related_to, location_of, adjacent_to, surrounds, traverses, functionally_related_to, affects, manages, treats, disrupts, complicates, interacts_with, prevents, brings_about, produces, causes, performs, carries_out, exhibits, practices, occurs_in, process_of, uses, manifestation_of, indicates, result_of, temporally_related_to, co-occurs_with, precedes, conceptually_related_to, evaluation_of, conceptual_part_of, issue_in, degree_of, analyzes, assesses_effect_of, measurement_of, measures, diagnoses, property_of, derivative_of, developmental_form_of, method_of, isa

Appendix B: Detailed templates

Here we present the detailed prompt/cypher templates mentioned but not included in the main text. Figure 5 is the prompt template for post-hoc explanation, Fig. 6 is the prompt template for self-assessment, which induces the model's unfamiliar concepts, Fig. 7 is the cypher query template of the retrieval of AnesGraph-RAG-UD, and Fig. 8 is the prompt used for question translation of the datasets.

```
===== Post-hoc Explanation =====
Knowledge = {Re_retrieval_graph_context}
Question = {question}
Reasoning process = {reasoning_process}
```

// Optional:

The current reasoning is based on eliminating incorrect options. Reframe the reasoning process as if ****no predefined choices are available****.

Constructing a clinically appropriate conclusion step by step from first principles. For example:

```
# Step 1: Identify the Clinical Problem and Urgency
# Step 2: Define the Immediate Therapeutic Goal
# Step 3: Enumerate Viable Therapeutic Categories Based on the Situation
# Step 4: Evaluate Each Category Against Clinical Constraints
# Step 5: Justify the Optimal Choice.
```

Please relate the key steps in the reasoning process to the Graph Nodes and Relationships, making the reasoning process explainable (using Source{xxx}).

Fig. 5 The prompt template for post-hoc explanation

```
===== Concept List Generation =====
```

You are a medical expert. Carefully extract ALL distinct and relevant medical concepts (such as diseases, treatments, metrics, symptoms, procedures, or medical terms) from the following question and its answer choices.

First, identify the major topic or main medical concepts of the question (if it exists), and add it to the front of the list (question-related concepts).

Then, for each answer choice, extract the most relevant medical concepts or terms. If a choice is not relevant, do not include it (choice-related concepts).

Lastly, list the concepts in a single line, with the major topic first, then the rest concepts from least familiar to you to most familiar to you (most unfamiliar first), separated by commas.

Please ensure that:

- Output only the concept list, with no extra explanation or formatting.
- Only include specific, meaningful medical terms or phrases directly related to the question context. Do not include general words or non-medical terms.
- **IMPORTANT:** For all terms in the list, use only standard, general medical terms (such as those found in ICD-10 or MeSH).
- DO NOT use overly specific, descriptive, or hard-to-match phrases (e.g., use 'asthma' instead of 'asthmatic attacks', 'hypoxemia' instead of 'severe hypoxemia', 'respiratory alkalosis' instead of 'acute respiratory alkalosis').
- Avoid using adjectives, qualifiers, or extra details—extract only the core medical concept. If unsure, always choose the more general or standard term.

Question: {question}

Fig. 6 The prompt template for self-assessment

```

MATCH (a)-[r]->(b)
WHERE
  // at least one concept is in node a.
  ANY(c IN $concepts WHERE
    toLower(a.description) CONTAINS c OR toLower(a.text)
  CONTAINS c
  )
  AND
  // at least one concept is in node b.
  ANY(c IN $concepts WHERE
    toLower(b.description) CONTAINS c OR toLower(b.text)
  CONTAINS c
  )
  AND
  // node a and b should contain all the concepts.
  ALL(c IN $concepts WHERE
    toLower(a.description) CONTAINS c OR toLower(a.text)
  CONTAINS c OR
    toLower(b.description) CONTAINS c OR toLower(b.text)
  CONTAINS c
  )
UNWIND [a, b] AS node
WITH DISTINCT node,
CASE
  WHEN '.__Entity__' IN labels(node) THEN node.description
  WHEN '.__Chunk__' IN labels(node) THEN node.text
  ELSE null
END AS content
RETURN content

```

Fig. 7 The cypher query template of AnesGraph-RAG-UD

```

===== System prompt =====
You are a translator, you translate the given Chinese text into
English text.

===== User message =====
The original text is: {text_to_translate}, and the translation
result is:

```

Fig. 8 The prompt used for question translation

Appendix C: Detailed ablation study

The full ablation study is demonstrated in Table 5, where the two additional rows are:

- **Ablation CoT:** Standard LLM + CoT. We use the same Chain-of-Thought reasoning example in the prompt as in AnesGraph-RAG, aiming to demonstrate that the retrieved context is essential for achieving the final accuracy, rather than the model being able to derive the answer solely through CoT reasoning.
- **Ablation Text-RAG:** Standard LLM + Text-RAG. This setup uses the Text-RAG method as a baseline for comparison.

Table 5 The ablation study of our AnesGraph-RAG method on three anesthesiology test sets using ChatGPT-4o as the base LLM. We detail a six-level ablation, starting from a standard LLM and incrementally adding our key components, text retrieval, graph retrieval, pre-judgment module, hybrid retrieval, and Chain-of-Thought (CoT) prompting, to arrive at the full AnesGraph-RAG model. The results demonstrate that each component positively contributes to performance

Model	Method	BK	PPS	PK	Average
ChatGPT-4o	Standard	72.00	63.50	70.50	68.67
	Ablation CoT	74.50	70.00	63.50	69.33
	Ablation Text-RAG	71.50	67.00	70.50	69.67
	Ablation 1	66.50	63.00	70.00	66.50
	Ablation 2	69.50	67.50	70.50	69.17
	Ablation 3	70.50	70.50	68.50	69.83
	Ablation 4	74.00	67.50	67.00	69.50
	AnesGraph-RAG	73.00	72.00	73.00	72.67

Appendix D: Supplementary information

For detailed prompts for the construction of the knowledge graph and the detailed question banks used to evaluate QA performance, we provide a separate supplementary file at <https://github.com/SYatSEU/HIS-Supplementary>.

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Data availability

The detailed question banks used in this study are publicly available in the GitHub repository [<https://github.com/SYatSEU/HIS-Supplementary/tree/main>].

Declarations

Conflict of interest

The authors declare that they have no conflict of interest.

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